Class 10 Science

Chapter 2
Acids, Bases and Salts

☆ QUICK MEMORY TRICK

"Acids are RED and SOUR → Bases are BLUE and BITTER"

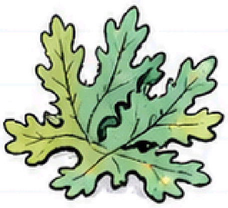
pH < 7 = Acid | pH = 7 = Neutral | pH > 7 = Base

1. INDICATORS — Types & Colour Changes

Indicator	In Acid	In Base
Blue Litmus	Turns Red	Stays Blue
Red Litmus	Stays Red	Turns Blue
Phenolphthalein	Colourless	Pink
Methyl Orange	Red	Yellow

Natural Indicators

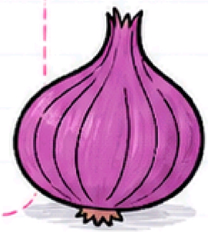
- Litmus (from lichen/Thallophyta) → purple when neutral
- Turmeric → turns reddish-brown with base
- Red cabbage, Hydrangea, Petunia, Geranium petals



Olfactory Indicators ☆

- Onion → smell disappears in base (NaOH), present in acid
- Vanilla → smell disappears in base
- Clove oil → smell disappears in base

Trick: Onion and clove = olfactory indicators.
Vanilla = also works.



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2. CHEMICAL REACTIONS OF ACIDS & BASES

Acid + Metal → Salt + Hydrogen gas ⭐

- $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$
- $Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$

Test for H_2: Burns with a pop sound

Base + Metal → Salt + Hydrogen gas ⭐

$2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2\uparrow$

(Sodium Zincate)

Note: Not all metals react with bases — only amphoteric metals like Zn, Al

Metal Oxide + Acid → Salt + Water

$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

(Blue-green colour)

Metal oxides are basic oxides

Non-metal Oxide + Base → Salt + Water

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

Non-metallic oxides are acidic in nature

Metal Carbonate/Bicarbonate + Acid → Salt + CO_2 + Water ⭐

- $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2\uparrow$
- $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2\uparrow$

Test for CO_2: Turns lime water milky

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$

On excess CO_2:

$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ (soluble – milkiness disappears)

Acid + Base → Salt + Water (Neutralisation) ⭐⭐

$NaOH + HCl \rightarrow NaCl + H_2O$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$

Key Points to Remember ⭐

- H_2 gas → Burns with a POP sound
- CO_2 gas → Turns lime water MILKY
- Metal oxides are BASIC oxides.
- Non-metallic oxides are ACIDIC in nature.
- Neutralisation → Acid + Base → Salt + Water.

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3. ACIDS & BASES IN WATER — KEY CONCEPT ★★

- Acids produce $H^+_{(aq)}$ / H_3O^+ in water
- Bases produce $OH^-_{(aq)}$ in water
- Dry HCl gas does NOT turn litmus red — needs water to ionise

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

$$H^+ + H_2O \rightarrow H_3O^+$$

(Hydronium ion)

Glucose & alcohol have H but NO H^+ ions in water → NOT acids

Dilution is Exothermic ⚠

- Always add ACID to WATER, never water to acid
- Adding water → decreases H_3O^+/OH^- concentration → called dilution

4. pH SCALE ★★★

0 ← 7 → 14

Very Acidic | Neutral | Very Alkaline (Base)

- pH stands for "potenz" (German = power)
- Measured using Universal Indicator (paper form)
- Higher H^+ concentration → Lower pH

pH in Daily Life (Board MCQ Favourites)

- Human body works pH 7.0 – 7.8
- Tooth decay begins pH < 5.5
- Acid rain pH < 5.6
- Stomach (HCl) ~ 1.2
- Blood pH 7.4
- Neutral pH = 7

Key Points to Remember ★

- Acids produce $H^+_{(aq)}$ / H_3O^+ in water.
- Bases produce $OH^-_{(aq)}$ in water.
- Dry HCl gas does NOT turn litmus red.
  → Needs water to ionise.
- Dilution means decreasing concentration of H_3O^+/OH^- ions.
- Higher H^+ concentration → Lower pH.

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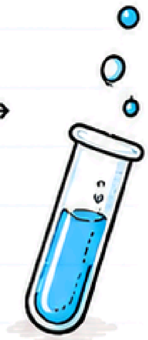
5. SALTS ⭐⭐

pH of Salts

Salt type	pH	Example
Strong acid + Strong base	= 7	NaCl
Strong acid + Weak base	< 7	NH_4Cl
Weak acid + Strong base	> 7	Na_2CO_3

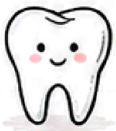
Family of Salts

- Same positive radical → e.g., NaCl, Na_2SO_4 = Sodium family
- Same negative radical → e.g., NaCl, KCl = Chloride family



Tooth Decay Mechanism ⭐

- Bacteria → degrades sugar → produces acid → pH drops below 5.5 → corrodes enamel (calcium hydroxyapatite)
- Remedy: Basic toothpaste neutralises the acid



Bee sting vs Wasp sting ⭐

- Bee sting = Acidic (methanoic acid) → apply baking soda (base)

- Wasp sting = Alkaline → apply vinegar (acid)

Key Points to Remember ⭐

- Salt formed from strong acid + strong base → pH = 7 (neutral).
- Salt formed from strong acid + weak base → pH < 7 (acidic).
- Salt formed from weak acid + strong base → pH > 7 (basic).
- Tooth decay occurs when pH falls below 5.5.
- Bee sting is acidic → use base (baking soda).
- Wasp sting is alkaline → use acid (vinegar).

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6. CHEMICALS FROM COMMON SALT ★★★

Chlor-Alkali Process

$$2NaCl_{(aq)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)}$$

- Anode → Cl_2 gas
- Cathode → H_2 gas + NaOH solution

Products & Uses (Must Know)

Product	Key Use
Cl_2	Water treatment, PVC, disinfectants
H_2	Fuel, margarine, fertilisers (ammonia)
NaOH	Soaps, detergents, paper, de-greasing

Bleaching Powder $Ca(ClO)_2$

$$2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$$

Uses: Bleaching textiles/wood pulp, disinfecting water, oxidising agent

Baking Soda ($NaHCO_3$) ★

$$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$$

On heating:

$$2NaHCO_3 \xrightarrow{heat} Na_2CO_3 + H_2O + CO_2$$

Uses: Baking powder (+ tartaric acid), antacids, fire extinguisher

Washing Soda ($Na_2CO_3 \cdot 10H_2O$)

$$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$$

Uses: Glass/soap/paper industry, removing permanent hardness of water, cleaning agent

Key Points to Remember ★

- Common salt (NaCl) gives us many useful chemicals.
- Chlor-alkali process produces Cl_2, H_2 and NaOH.
- Bleaching powder is an oxidising and disinfecting agent.
- Baking soda is also used in antacids and baking powder.
- Washing soda removes permanent hardness of water.

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7. WATER OF CRYSTALLISATION ☆☆

Definition: Fixed number of water molecules present in one formula unit of a salt.

Salt	Formula	No. of Water Molecules
Copper sulphate (Blue vitriol)	$CuSO_4 \cdot 5H_2O$	5
Washing soda	$Na_2CO_3 \cdot 10H_2O$	10
Gypsum	$CaSO_4 \cdot 2H_2O$	2
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$	½

☆ Anhydrous $CuSO_4$ is WHITE; add water to it → turns BLUE again.

Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$) ☆☆

Preparation from Gypsum

$$\underset{\text{(Gypsum)}}{CaSO_4 \cdot 2H_2O} \xrightarrow{373\ K} \underset{\text{(Plaster of Paris)}}{CaSO_4 \cdot \tfrac{1}{2}H_2O} + 1\tfrac{1}{2}H_2O$$

Setting Hard (Reverse Reaction)

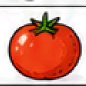
$$CaSO_4 \cdot \tfrac{1}{2}H_2O + 1\tfrac{1}{2}H_2O \longrightarrow \underset{\text{(Hard mass)}}{CaSO_4 \cdot 2H_2O}$$

Uses

- Used for casts of fractured bones
- Making toys, decorative materials
- Used to get smooth surfaces

☆ Store in moisture-proof container – absorbs moisture and sets hard.

8. NATURALLY OCCURRING ACIDS ☆☆

Source	Acid
Vinegar	Acetic acid 
Lemon / Orange	Citric acid 
Tamarind	Tartaric acid 
Tomato	Oxalic acid 
Curd	Lactic acid 
Ant / Nettle sting	Methanoic acid 

9. ANSWER TIPS (BOARD FAVOURITES) ☆☆

1. **Why acid to water, not water to acid?**
   → Dissolving acid in water is exothermic. Adding water to acid causes sudden heat, mixture splashes and can cause burns.
2. **Why dry HCl does not show acidic properties?**
   → H^+ ions form only in presence of water. Dry HCl cannot ionise.
3. **Why distilled water doesn't conduct but rain water does?**
   → Rain water dissolves CO_2 → forms carbonic acid → produces ions → conducts. Distilled water has no ions.
4. **Why glucose/alcohol not acids despite having H?**
   → They don't produce H^+ ions in water — shown by electrical conductivity test (bulb doesn't glow).
5. **Why $NaHCO_3$ used as antacid?**
   → Mild alkaline salt, neutralises excess HCl in stomach without being corrosive.

Key Points to Remember ☆

- Water of crystallisation = Fixed number of water molecules in a salt.
- Gypsum ($CaSO_4 \cdot 2H_2O$) → Heat → Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$).
- Plaster of Paris + water → Gypsum (hard mass).
- Common salt (NaCl) is the raw material for many useful chemicals.
- Naturally occurring acids are found in our daily life.